LPP

points to remember in graphical method

* existence of optimal soln. for an LPP may not always exist an corner pts. It depends on the objective func.
* 2 pt. A & B have same optimal soln. Then LPP has ∞'ly many soln. in line segment AB

points for simplex problem

* coeff corresponding to surplus/slack variable is set to zero
* for a variable x unrestricted in sign write it as

$$x' - x''$$

* non basic variable → variable equated to zero

n = # variables , m = # equation

non basic variable $= n - m$

basic variable → variable not equated to zero

basic soln. $= \binom{n}{m}$

basic feasible solution → soln where basic variable are non negative

non degenerate basic feasible soln → soln. where basic variable are strictly +ve

optimality condition

for minimization z row must be non +ve

maximization z row must be non -ve

points for big M method

- when constraints in form $\geq$ or $=$ then the initial feasible soln. is not easily available

for a case $ax \geq -b$ → don't use big M take -ve on both sides

$ax \geq b$

↳ use big M method

- artificial variable not part of final soln. to ensure this we add a penalty M (minimization) or $-M$ (maximization) as a coeff for all artificial variables in obj. fn Z

- when an artificial variable is a leaving variable the column corresponding to the artificial variable must be removed.

Shortcuts to remember

for graphical method

i) for finding intercepts of constraints

$$ax_1 + bx_2 = c$$

$$\frac{x_1}{c/a} + \frac{x_2}{c/b} = 1 \Rightarrow (c/a, 0) \& (0, c/b)$$ are the intercepts

ii) for finding intersection btw. lines use calc. to solve the equations.

for simplex method

i) always set algebraic variables as zero when finding basic feasible soln.

ii) instead of performing row operations you could perform

b a
c d
↑

if c is pivot element and we need to find new value of a

$$a_{new} = \frac{ca - bd}{c}$$

note always the row ← then elements corresponding to pivot column must be multiplied with -1

iii) when there is a minimization problem convert it to maximization by calculation $Z^* = -Z$

for big M method

i)

1.

	A	B	min req.
V_1	1	1	1
V_2	100	10	50
V_3	10	100	10
Cost	1	1.5	

$x_1 \rightarrow$ units of food A consumed

$x_2 \rightarrow$ units of food B consumed

V_1 Consumed $= x_1 + x_2 \geq 1$

V_2 Consumed $= 100x_1 + 10x_2 \geq 50$

V_3 Consumed $= 10x_1 + 100x_2 \geq 10$

daily cost $= x_1 + 1.5x_2 \rightarrow$ minimize.

LPP

$$\min Z = x_1 + 1.5x_2$$

subject to $x_1 + x_2 \geq 1$

$100x_1 + 10x_2 \geq 50$

$10x_1 + 100x_2 \geq 10$